

- Let $f(x, y) = 0$ be the equation of a circle. If $f(0, \lambda) = 0$ has equal roots $\lambda = 2, 2$ and $f(\lambda, 0) = 0$ has roots $\lambda = \frac{4}{5}, 5$, then the centre of the circle is
 (a) $\left(2, \frac{29}{10}\right)$ (b) $\left(\frac{29}{10}, 2\right)$ (c) $\left(-2, \frac{29}{10}\right)$ (d) None
- The polar of the point $\left(5, -\frac{1}{2}\right)$ with respect to circle $(x-2)^2 + y^2 = 4$ is
 (a) $5x - 10y + 2 = 0$ (b) $6x - y - 20 = 0$
 (c) $10x - y - 10 = 0$ (d) $x - 10y - 2 = 0$
- If circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch each other, then
 (a) $\frac{1}{a} + \frac{1}{b} = \frac{1}{c}$ (b) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$
 (c) $\frac{1}{a} + \frac{1}{b} = c^2$ (d) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c}$
- The common chord of $x^2 + y^2 - 4x - 4y = 0$ and $x^2 + y^2 = 16$ subtends at the origin an angle equal to
 (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$
- If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point on the y-axis, then the length of PQ is
 (a) 4 (b) $2\sqrt{5}$ (c) 5 (d) $3\sqrt{5}$
- If length of common chord of two circles $x^2 + y^2 + 8x + 1 = 0$ and $x^2 + y^2 + 2\mu y - 1 = 0$ is $2\sqrt{6}$, then value of μ are
 (a) ± 2 (b) ± 3
 (c) ± 4 (d) None of these
- The range of parameter 'a' for which the variable line $y = 2x + a$ lies between the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle, is -
 (a) $(-15 + 2\sqrt{5}, -\sqrt{5} - 1)$ (b) $(15 + 2\sqrt{5}, \sqrt{5} - 1)$
 (c) $(-15 - 2\sqrt{5}, -\sqrt{5} + 1)$ (d) $(-15 + 2\sqrt{5}, \sqrt{5} - 1)$
- If equation $(a+1)x^2 + (b-1)y^2 + (a-2b)xy - 6x + 9y - 3 = 0$ represent a circle then its radius is
 (a) $\frac{3}{2}$ (b) $\frac{2}{3}$
 (c) 1 (d) $\sqrt{a^2 + b^2}$
- If a circle of constant radius $3k$ passes through the origin and meets the axes at A and B, the locus of centroid of ΔOAB is-
 (a) $x^2 + y^2 = k^2$ (b) $x^2 + y^2 = 2k^2$
 (c) $x^2 + y^2 = 3k^2$ (d) $x^2 + y^2 = 4k^2$
- If the circle $x^2 + y^2 - 4x - 6y + \lambda = 0$ touches x-axis then the value of λ is
 (a) -4 (b) 4 (c) -8 (d) 8
- The equation of the line parallel to the line $3x + 4y = 0$ and touching the circle $x^2 + y^2 = 9$ in the first quadrant is -
 (a) $3x + 4y = 9$ (b) $3x + 4y = 45$
 (c) $3x + 4y = 15$ (d) None of these
- The locus of the centre of a circle which touches externally the circle $x^2 + y^2 - 6x - 6y + 14 = 0$ and also touches the y-axis is given by the equation -
 (a) $x^2 - 6x - 10y + 14 = 0$ (b) $x^2 - 10x - 6y + 14 = 0$

- (c) $y^2 - 6x - 10y + 14 = 0$ (d) $y^2 - 10x - 6y + 14 = 0$
13. The circle $x^2 + y^2 - 6x - 10y + p = 0$ does not touch nor intersect the coordinate axes if the point (1, 4) lies inside the circle then -
(a) $0 < p < 29$ (b) $25 < p < 29$
(c) $25 < p < 35$ (d) None of these
14. The equation of the diameter of the circle $x^2 + y^2 - 6x + 2y - 8 = 0$ is given by
(a) $x + 3y = 0$ (b) $x - 3y = 0$
(c) $x - 2y = 0$ (d) $x + 2y = 0$
15. If the chord $y = mx + 1$ of the circle $x^2 + y^2 = 1$ subtends an angle of measure 45° at the major segment of the circle then value of m is
(a) 2 (b) -2 (c) ± 1 (d) None of these
16. If the distances from the origin to the centres of three circles $x^2 + y^2 + 2\lambda_i x - c^2 = 0$ ($i = 1, 2, 3$) are in G.P. then the lengths of the tangents drawn to them from any point on the circle $x^2 + y^2 = c^2$ are in
(a) A.P. (b) G.P. (c) H.P. (d) None of these
17. A foot of the normal from the point (4, 3) to a circle is (2, 1) and a diameter of the circle has the equation $2x - y = 2$. Then the equation of the circle is
(a) $x^2 + y^2 + 2x - 1 = 0$ (b) $x^2 + y^2 - 2x - 1 = 0$
(c) $x^2 + y^2 - 2y - 1 = 0$ (d) None of these
18. The number of integral value of λ for which $x^2 + y^2 + 2\lambda x + 2(1 - \lambda)y - 1 = 0$ is the equation of a circle whose radius can not exceed 3
(a) 3 (b) 4 (c) 5 (d) None
19. In a triangle ABC, $r^2 + r_1^2 + r_2^2 + r_3^2 + a^2 + b^2 + c^2$ is equal to -
(a) $4R^2$ (b) $8R^2$ (c) $12R^2$ (d) $16R^2$
20. The area of the circle and the area of a regular polygon of n sides, and of perimeter equal to that of the circle are in the ratio of

- (a) $\tan\left(\frac{\pi}{3}\right) : \frac{\pi}{n}$ (b) $\cot\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$
(c) $\sin\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$ (d) $\cos\left(\frac{\pi}{n}\right) : \frac{\pi}{n}$